

SQL Commands Reference: DDL, DML & TCL

A Concise, Structured & Colorful Cheat Sheet for Database Management Systems

1. DDL (Data Definition Language)

STRUCTURE & SCHEMA

- **Primary Scope:** Defines, alters, and manages the overall **schema or structural framework** of database objects (tables, views, indexes).
- **Auto-Committed:** DDL operations are automatically saved permanently to the database immediately upon execution and **cannot be rolled back**.

Command	Purpose	SQL Syntax & Example
CREATE	Creates a new database object (table, view, index).	CREATE TABLE Students (ID INT, Name VARCHAR(50));
ALTER	Modifies the existing table structure (adds/removes columns).	ALTER TABLE Students ADD Age INT;
DROP	Permanently deletes a table along with its data and schema.	DROP TABLE Students;
TRUNCATE	Deletes all rows from a table while preserving its structure.	TRUNCATE TABLE Students;
RENAME	Renames an existing database object.	RENAME TABLE Students TO RegularStudents;

Example DDL Execution Flow:

```
-- 1. Create a new table structure
CREATE TABLE Employees (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(100),
    Salary DECIMAL(10, 2)
);

-- 2. Modify the table structure
ALTER TABLE Employees ADD Department VARCHAR(50);
```

2. DML (Data Manipulation Language)

DATA OPERATIONS

- **Primary Scope:** Used for inserting, modifying, deleting, and retrieving **actual data records** stored within existing tables.
- **Not Auto-Committed:** Changes made via DML commands can be **rolled back** or committed using Transaction Control commands.

Command	Purpose	SQL Syntax & Example
INSERT	Adds new rows/records into a table.	INSERT INTO Employees VALUES (101, 'Anand', 50000, 'IT');
UPDATE	Modifies existing data records based on a condition.	UPDATE Employees SET Salary = 55000 WHERE EmpID = 101;
DELETE	Removes specific rows matching a conditional clause.	DELETE FROM Employees WHERE EmpID = 101;

Comparative Analysis: DROP vs TRUNCATE vs DELETE

Command	Type	Target / Operation Scope	Rollback Status
DROP	DDL	Removes entire table (Schema + Data)	Cannot Rollback
TRUNCATE	DDL	Deletes all rows rapidly (Preserves Schema)	Cannot Rollback
DELETE	DML	Deletes specific or all matching rows	Can Rollback

3. TCL (Transaction Control Language)

TRANSACTION SAFETY

- **Primary Scope:** Manages transactions to ensure data consistency and integrity across multiple DML operations.
- **Transaction Concept:** A logical unit of work consisting of one or more sequential DML commands executed together.

Command	Purpose	SQL Syntax & Example
COMMIT	Permanently saves all transaction changes to the database.	COMMIT;
ROLLBACK	Reverts all changes made since the last commit or savepoint.	ROLLBACK;
SAVEPOINT	Establishes a intermediate checkpoint within a transaction.	SAVEPOINT SavePointA;

Step-by-Step Transaction Workflow:

```
-- Step 1: Insert first employee record
INSERT INTO Employees VALUES (102, 'Kavitha', 60000, 'HR');

-- Step 2: Establish a checkpoint
SAVEPOINT Checkpoint1;

-- Step 3: Insert second employee record
INSERT INTO Employees VALUES (103, 'Suresh', 45000, 'Finance');

-- Step 4: Revert back to Checkpoint1 (Undoes Step 3 only)
ROLLBACK TO Checkpoint1;

-- Step 5: Save changes permanently
COMMIT;
```